A scientist sequences the genomes of several species known to be evolutionarily related to humans. The percentage of similarity between the genome of each species to the genome of humans is calculated and shown in the table below.

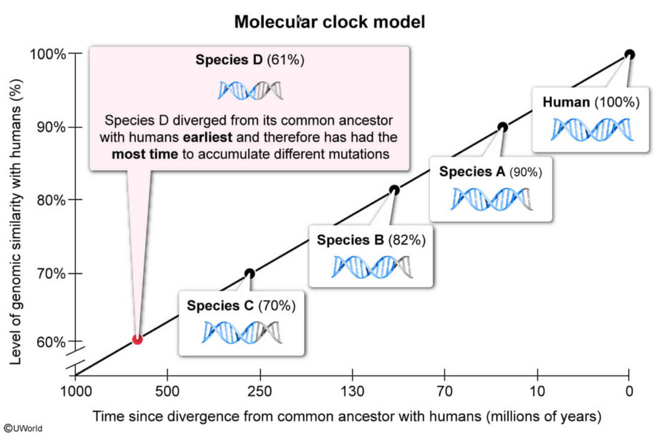
	Genomic similarity to humans
Species A	90%
Species B	82%
Species C	70%
Species D	61%

Given this, which of the following statements regarding these species is most likely true?

- A. Species A diverged from the last common ancestor it shared with humans earlier than any of the other species. [13%]
- B. Species B and humans diverged from a common ancestor more recently than Species A and humans. [1%]
- C. Species C and humans diverged from a common ancestor less recently than Species D and humans. [1%]
- D. Species D diverged from the last common ancestor it shared with humans earlier than any of the other species. [83%]

Correct
83% answered correctly

Explanation:



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Based on the **neutral theory of molecular evolution**, the majority of genetic mutations are neutral, which means that they do not affect the fitness of an organism. Within species, these **neutral mutations** are randomly **fixed or lost** due to genetic drift and occur at a **fairly constant rate** over time. After

two species diverge (separate), the *accumulation rate* of neutral mutations in a region of homologous DNA is mostly similar in both species. Accordingly, this "molecular clock" of evolution is said to be linear because the number of neutral mutations generally **increases** with time.

Therefore, the molecular clock model allows researchers to measure evolutionary time by analyzing random changes in the genome over time. For example, species that diverged *more recently* from a common ancestor have accumulated fewer mutations across a shorter time and are therefore *more* genetically similar to one another (ie, more closely related). In contrast, species that diverged *less recently* (eg, longer ago) from a common ancestor have accumulated more mutations across a longer period and are *less* genetically similar to one another (ie, less closely related).

In this scenario, a scientist sequences the genomes of several species that are evolutionarily related to humans and calculates the percentage of similarity between the genome of each species and the human genome. According to the table, the **Species D** genome is **leastsimilar** to the human genome (**61%** similarity). This indicates that, compared with the other species, Species D (*not* Species A) likely diverged from its last common ancestor with humans **earliest** in evolutionary time and has had the *most* time to accumulate mutations *different* from those in humans **(Choice A)**.

(Choice B) The Species B genome is *less* similar to the human genome (82%) than the Species A genome is to the human genome (90%). Therefore, Species B and humans likely diverged from a common ancestor *less* recently than Species A and humans (ie, Species B and humans are *less* closely related).

(Choice C) The Species C genome and human genome are *more* similar (70%) than the Species D genome and human genome (61%). Therefore, Species C and humans likely diverged from a common ancestor *more* recently than Species D and humans (ie, Species C and humans are *more* closely related).

Educational objective:

The molecular clock model is based on the theory that most genetic mutations are neutral and occur at a fairly constant rate across organisms. The model helps estimate the amount of time elapsed since species diverged from their common ancestor.

Time spent: 0 sec
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